Roadmap. Srednicki §2: how states and fields transform, the basis of every Lorentz-invariant theory. pp. 1–6: the group from Eq. 0; $\det\Lambda$ and the sign of $\Lambda^0{}_0$ split it into four components (p. 13, read after p. 6). pp. 7–11: $U(\Lambda)$, $M^{\mu\nu}$, Lorentz and Poincaré algebras. p. 12: $U^{-1}\varphi U = \varphi(\Lambda^{-1}x)$. Used in LecNote3, LecNote4 p. 7, LecNote16 (§33); P, T: §23. Supplements: LorentzGroup, FieldTransformations, GroupTheoryIntro ($+---$ metric).

Sec. 2.

Lorentz Invariance & Transformations.

(It is a non-compact continuous group, which creates a lot of additional works and non-Abelian & notorious)

A Lorentz transformation is linear, homogeneous change of coordinates

$$\overline{x}^\mu = \Lambda^\mu{}_\nu x^\nu$$

(see my lecture notes in the "additional material" for an introduction to representations & group theory in particle physics)

distance preserved:

$$x^2 \equiv x^\mu x_\mu = g_{\mu\nu} x^\mu x^\nu$$
$$= x^2 - c^2t^2$$

(Lie groups & their Exponential Representation).

and the metric obeys.

$$g_{\mu\nu}\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = g_{\rho\sigma} \quad \ldots \boxed{\text{Eq. 0}}$$

$$g_{\mu\nu} = \begin{pmatrix} -1 & & & \\ & +1 & & \\ & & +1 & \\ & & & +1 \end{pmatrix}$$ Minkowski metric

for $\Lambda = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & R_{ij} & \\ 0 & & & \end{pmatrix}$ ← ordinary rotations [1]

[1] Should read $\Lambda = \mathrm{diag}(1, R_{ij})$ with $R \in SO(3)$: $\Lambda^0{}_0 = 1$. With 0 there, Eq. 0 at $\rho = \sigma = 0$ gives $0 = -1$, and $\det\Lambda = 0$.

All Lorentz transformations form a **Group**.

Product is associative: two consecutive L.T. still a L.T.
(2 closure)

Identity exist: $\exists\ \Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu}$

Inverse exist: $\exists\ \Lambda^{-1}$ that $(\Lambda^{-1})^{\mu}{}_{\nu}\Lambda^{\nu}{}_{\sigma} = \delta^{\mu}{}_{\sigma}$

~~And closure~~: (Let's explore a few familar properties in the vector Rep of Lorentz Group, sometimes we call it **defining representation**)

$$g_{\mu\nu}\Lambda^{\mu}{}_{\rho}\Lambda^{\nu}{}_{\sigma} = g_{\rho\sigma}$$

$g^{\rho\alpha}$ $\quad$ $g^{\rho\alpha}$

$$\Lambda g_{\nu\rho}\Lambda^{\nu}{}_{\sigma}\, g^{\rho\alpha} = g_{\rho\sigma}\, g^{\rho\alpha}$$

$$\Lambda\nu^{\alpha}\Lambda^{\nu}{}_{\sigma} = \delta_{\sigma}{}^{\alpha} = \delta^{\alpha}{}_{\sigma} \quad \cdots (A)$$

we would also demand:

$$(\Lambda^{-1})^{\alpha}{}_{\cdot\nu}\Lambda^{\nu}{}_{\sigma} = \delta^{\alpha}{}_{\sigma} \quad \cdots (B)$$

such that we know (compare (A) and (B))

$$(\Lambda^{-1})^{\alpha}{}_{\nu} = \Lambda_{\nu}{}^{\alpha}$$

we also have

$$g^{\mu\nu}\Lambda\rho_{\mu}\Lambda^{\sigma}{}_{\nu} = g^{\rho\sigma} \quad \cdots$$ prove it using the above

For infinitesimal transformation.

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \delta\omega^{\mu}{}_{\nu}.$$

From Eq.0 we can see that

$$\delta\omega_{\rho\sigma} = -\delta\omega_{\sigma\rho}$$

$$\delta\omega^{\mu\nu} = -\delta\omega^{\mu\nu} \quad \cdots \text{(H.W.)}.$$ [1]

Hence in 4D, there are 6 independent

L.T.s (why $\delta\omega$ real? for students)
coordinates are real).

6 L.T.s
- } 3 rotations. $\delta\omega_{ij} = -\epsilon_{ijk}\hat{n}_k\delta\theta.$
- } 3 boosts $\delta\omega_{i0} = \hat{n}_i\delta\eta.$

☆ Not All L.T.s can be done through compounding infinitesimal ones:

back to the basics of Λ

note that the core is about their product.

Hence sign is untixed, e.g.,

$$(\Lambda^{-1})^{\mu}{}_{\sigma} = \Lambda_{\sigma}{}^{\mu}$$

$$\mathrm{Det}(\Lambda^{-1}) = \mathrm{Det}(\Lambda).$$
$$= (\mathrm{Det}\Lambda)^{-1}$$

1 Should read $\delta\omega^{\mu\nu} = -\delta\omega^{\nu\mu}$: raise both indices of the line above. As written it forces $\delta\omega^{\mu\nu} = 0$. The mixed $\delta\omega^{\mu}{}_{\nu}$ is not antisymmetric: $\delta\omega^{0}{}_{i} = \delta\omega^{i}{}_{0}$.

it implies $\det \Lambda = \pm 1$.

$\Lambda \rightarrow \det \Lambda = 1$ proper

$\Lambda \rightarrow \det \Lambda = -1$ improper.

Note that the product of any two L.T.s are **proper**.[1]

And any infinitesimal L.T.s are proper, as

$$\Lambda = 1 + \delta w.$$

Any L.T.s can be reached by compounding[2] infinitesimal L.T.s are proper.

these forms a subgroup of L.T.s.

Yet another subgroup is **orthochronous** L.Ts.

those for which $\Lambda^0{}_0 \geq +1$.

Eq (0) implies (take $\rho = \sigma = 0$)

$$(\Lambda^0{}_0)^2 - (\Lambda^1{}_0 \Lambda^1{}_0) = 1.$$

$$\Rightarrow \Lambda^0{}_0 \geq +1 \text{ or } \Lambda^0{}_0 \leq -1.$$

1 Should read: the product of two proper (or of two improper) LTs is proper, since $\det(\Lambda'\Lambda) = \det\Lambda' \det\Lambda$; P times a proper LT is improper. Hence proper LTs form a subgroup.

2 Read: any LT that can be reached by compounding is proper. Why p. 3's starred claim holds: $\det\Lambda = \pm 1$ exactly, and along a continuous path of LTs it cannot jump; nor can $\Lambda^0{}_0$ cross $(-1, 1)$ (below). Paths from 1 keep $\det\Lambda = 1$, $\Lambda^0{}_0 \geq 1$, so P, T and even the proper $PT = -1$ are out of reach.

Clearly infinitesimal L.T.s are ~~chronous~~ orthochronous and the product of two L.T.s ~~are~~ orthochronous ones are orthochronous.[1]

thus. Proper & Orthochronous L.T.s forms a subgroup via a series of infinitesimal L.T.s.

Two discrete operations can take us out of this subgroups: Parity & Time-Reversal.

Parity $\mathcal{P}^{\mu}{}_{\nu} = (\mathcal{P}^{-1})^{\mu}{}_{\nu} = \begin{pmatrix} +1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$.

proper $\longleftrightarrow$ improper

remains ~~chronous~~ orthochronous

1 Not obvious: $(\Lambda'\Lambda)^0{}_0 = \Lambda'^0{}_0\Lambda^0{}_0 + \Lambda'^0{}_i\Lambda^i{}_0$. Put $\Lambda^0{}_0 = \cosh\alpha$; then the column $\Lambda^i{}_0$ has length $\sinh\alpha$ (p. 4) and the row $\Lambda'^0{}_i$ length $\sinh\alpha'$ (p. 2's $g^{\mu\nu}$ identity), so by Cauchy–Schwarz $(\Lambda'\Lambda)^0{}_0 \geq \cosh(\alpha' - \alpha) \geq 1$. Conversely, every proper orthochronous Λ is a boost times a rotation, both reached by compounding.

Time Reversal:

$$\mathcal{T}^{\mu}{}_{\nu} = (\mathcal{T}^{-1})^{\mu}{}_{\nu} = \begin{pmatrix} -1 & & & \\ & +1 & & \\ & & +1 & \\ & & & +1 \end{pmatrix}.$$

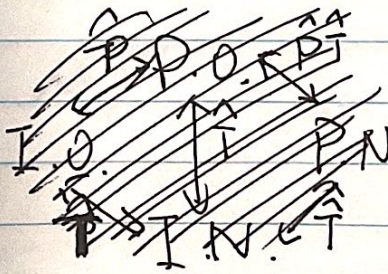
proper $\longleftrightarrow$ improper.

orthochronous $\longleftrightarrow$ non-orthochronous.

Note: Generally, when we say a theory is Lorentz invariant, we mean invariant under | proper orthochronous | subgroup of Lorentz.

A theory can be not invariant under. ~~P & T.~~ P / T.

Now let's focus on Proper Orthochronous subgroup. and revisit P/T in the ~~near~~ future.



(We'll see in this language $\Lambda^\mu{}_\nu$ is nothing but a vector representation, and $\delta\omega$ in here means angle.) (There is a mix of notation in the textbook,)

In quantum theory, symmetries are represented by unitary (anti-unitary) transformations.

(Now, a general representation)

→ We can associate a unitary operator to

~~$U^\dagger(\Lambda)U(\Lambda) = I$~~ norm being one

each proper orthochronous L.T. Λ.

these operators must obey the composition rule:

$$U(\Lambda'\Lambda) = U(\Lambda')\,U(\Lambda).$$

For an infinitesimal transformation, we can write.

$$U(1+\delta\omega) = I + \frac{i}{2}\delta\omega_{\mu\nu}M^{\mu\nu}$$

where $M^{\mu\nu}$ (symmetry factor) is a set of hermitian operators (we will see the meaning later). [1]

called the Generators of the Lorentz Group

$$M^{\mu\nu} = -M^{\nu\mu}$$

[1] The $\frac{1}{2}$: $\delta\omega_{\mu\nu}M^{\mu\nu}$ counts each pair twice. The i: to first order $U^\dagger U = I + \frac{i}{2\hbar}\delta\omega_{\mu\nu}(M^{\mu\nu} - M^{\mu\nu\dagger})$, so real $\delta\omega$ (p. 3) and unitary U need hermitian M. The $\hbar$: with p. 3's $\delta\omega$ and p. 10's J, K, $U = I - \frac{i}{\hbar}\delta\theta\,\hat{n}\cdot\mathbf{J} + \frac{i}{\hbar}\delta\eta\,\hat{n}\cdot\mathbf{K}$, and $e^{-i\theta\hat{n}\cdot\mathbf{J}/\hbar}$ is QM's rotation.

Now let's explore/motivate some properties of generators $M^{\mu\nu}$

Try $U(\Lambda^{-1})U(\Lambda')U(\Lambda) = U(\Lambda^{-1}\Lambda'\Lambda)$.

let $\Lambda' = 1 + \delta\omega'$ and expand both side by $\delta\omega'$ in linear order.

$$U(\Lambda)^{-1}\delta\omega'_{\mu\nu} M^{\mu\nu} U(\Lambda)$$

$$= U(\cancel{\Lambda^{-1}\Lambda\Lambda}\,(\Lambda^{-1})^{\mu}{}_{\nu}\Lambda'^{\nu}{}_{\rho}\Lambda^{\rho}{}_{\sigma})\Big|_{\text{linear}}$$

$$= U(\Lambda\nu^{\mu}(1+\delta\omega'^{\nu}{}_{\rho})\Lambda^{\rho}{}_{\sigma})\Big|_{\text{linear}}$$

$$= U(1+\delta\omega'^{\nu}{}_{\rho}\Lambda\nu^{\mu}\Lambda^{\rho}{}_{\sigma})\Big|_{\text{linear}}.$$

$$\cancel{= \delta\omega'^{\mu\nu}{}_{\rho}\Lambda\nu^{\mu}\Lambda^{\rho}{}_{\sigma}\Lambda^{\alpha\sigma}g^{\alpha\sigma}\delta^{\mu}{}_{\sigma}M}$$

$$= U(1+(\delta\omega'\Lambda_{\text{I}}\Lambda)^{\mu}{}_{\sigma})\Big|_{\text{linear}}$$

$$= U(1+(\delta\omega'\Lambda\Lambda)_{\rho\sigma}g^{\mu\rho})\Big|_{\text{linear}}.$$

$$= I + \delta\omega'_{\mu\nu}\Lambda^{\mu}{}_{\rho}\Lambda^{\nu}{}_{\sigma}M^{\rho\sigma} \quad \text{H.W.}$$

Since $\delta\omega'_{\mu\nu}$ is arbitrary.

$$\underbrace{U(\Lambda^{-1})M^{\mu\nu}U(\Lambda)}_{(\text{LHS}-I)/\delta\omega'_{\mu\nu}} = \Lambda^{\mu}{}_{\rho}\Lambda^{\nu}{}_{\sigma}M^{\rho\sigma}$$

clearly $U(\Lambda^{\mu})^{-1} M^{\mu\nu} \not\propto U(\Lambda)$

has each vector index μ, ν goes through its own Lorentz transformation.

This is general: any operator carries Lorentz index should transform accordingly.

For instance $\hat{p}^{\mu} = (\hat{H}, \hat{p}^{i})$

$$c\hat{p}^{0} = \hat{H}$$

$$U(\Lambda)\hat{p}^{\mu} U(\Lambda) = \Lambda^{\mu}{}_{\nu} \hat{p}^{\nu} \quad \cdots \quad (2.15)$$

Now if we let $\Lambda = 1 + \delta\omega$, expand to linear order in $\delta\omega$, we get the commutation relation, "Lie Algebra" of the Lorentz group.

$$[M^{\mu\nu}, M^{\rho\sigma}] = i\hbar(g^{\mu\rho} M^{\nu\sigma} - (\mu \leftrightarrow \nu)) - (\rho \leftrightarrow \sigma).$$

$$= i\hbar [g^{\mu\rho} M^{\nu\sigma} - g^{\nu\rho} M^{\mu\sigma} - g^{\mu\sigma} M^{\nu\rho} + g^{\nu\sigma} M^{\mu\rho}]$$ [1]

[1] Put $\Lambda = 1 + \delta\omega$ in p. 8's result: the left side is $M^{\mu\nu} + \frac{i}{2\hbar}\delta\omega_{\rho\sigma}[M^{\mu\nu}, M^{\rho\sigma}]$, the right $M^{\mu\nu} + \delta\omega^{\mu}{}_{\rho} M^{\rho\nu} + \delta\omega^{\nu}{}_{\rho} M^{\mu\rho}$. Write $\delta\omega^{\mu}{}_{\rho} M^{\rho\nu} = \frac{1}{2}\delta\omega_{\alpha\rho}(g^{\mu\alpha} M^{\rho\nu} - g^{\mu\rho} M^{\alpha\nu})$, likewise for ν, and match the coefficients of $\delta\omega_{\alpha\rho}$, both antisymmetric. (2.15) gives p. 10's $[P^{\mu}, M^{\rho\sigma}]$ the same way.

We can identify (to a representation) that.

Components of the angular momentum operator $\vec{J}$ as $J_i \equiv \frac{1}{2}\epsilon_{ijk} M^{jk}$.

components of the boost operator $\vec{K}$ as $K_i \equiv M^{i0}$

Then from the Lie Algebra, we have

$[J_i, J_j] = i\hbar \epsilon_{ijk} J_k$. ← commutation of angular momentum

$[J_i, K_j] = i\hbar \epsilon_{ijk} K_k$. ← K transforms as a 3-vector under rotation

$[K_i, K_j] = -i\hbar \epsilon_{ijk} J_k$. ← a series of boost can be related to rotation.

(This also is a reflection of the non-Abelian nature)

Similarly, we can show that. (from Eq 2.15)

$[P^\mu, M^{\rho\sigma}] = i\hbar(g^{\mu\sigma}P^\rho - (\rho \leftrightarrow \sigma))$.

(Note that translation generators do not commute with Lorentz generators, so Poincare group is not simple outter product of translation ⊗ Lorentz)

which gives.

$[J_i, H] = 0$.

$[J_i, P_j] = i\hbar \epsilon_{ijk} P_k$.

$[K_i, H] = i\hbar c P_i$

$[K_i, P_j] = i\hbar \delta_{ij} H/c$.

also $[P_i, P_j] = 0$

$[P_i, H] = 0$.

Together, they formed the Poincaré group (with translation).

Let's now consider what happens for a scalar field $\varphi(x)$.

under a Lorentz transformation:

back to the Heisenberg picture.

$$e^{+iHt/\hbar}\,\varphi(x,0)\,e^{-iHt/\hbar} = \varphi(x,t).$$

↓ Relativistic

$$e^{-ip\cdot x/\hbar}\,\varphi(0)\,e^{ipx/\hbar} = \varphi(x).$$

$$px = p^\mu x_\mu = \vec{p}\cdot\vec{x} - Ht.$$

We can make it a bit fancier by defining

unitary space-time translation operator

$$T(a) \equiv e^{-ip^\mu a_\mu/\hbar}.$$

Then we have

$$T(a)^{-1}\varphi(x)T(a) = \varphi(x-a) \quad \ldots \quad (2.24)$$

(new coordinate; relates to the other one.)

For an infinitesimal transformation.

$$T(\delta a) = I - \frac{i}{\hbar}\delta a_\mu P^\mu.$$

(how depends on metric choice; we do want to keep $\frac{i}{\hbar}\delta a_\mu$.)

Comparing with (Eq. 2.12) $\quad U(1+\delta\omega) = I + \frac{i}{2\hbar}\delta\omega_{\mu\nu}M^{\mu\nu}$

we see $U(\Lambda)^{-1}\varphi(x)U(\Lambda)$ in (2.24) means.

$$U(\Lambda)^{-1}\varphi(x)U(\Lambda) = \varphi(\Lambda^{-1}x). \quad (1)$$

Derivatives behave following the rules.

$$U(\Lambda)^{-1}\partial^\mu\varphi(x)U(\Lambda) = \Lambda^\mu{}_\nu\bar{\partial}^\nu\varphi(\Lambda^{-1}x).$$

and hence

$$U(\Lambda)^{-1}\partial^2\varphi(x)U(\Lambda) = \bar{\partial}^2\varphi(\Lambda^{-1}x).$$

so that

$$U(\Lambda)^{-1}\left(-\partial^2\varphi(x) + m^2/\hbar^2 c^2\varphi(x) = 0\right)U(\Lambda)$$

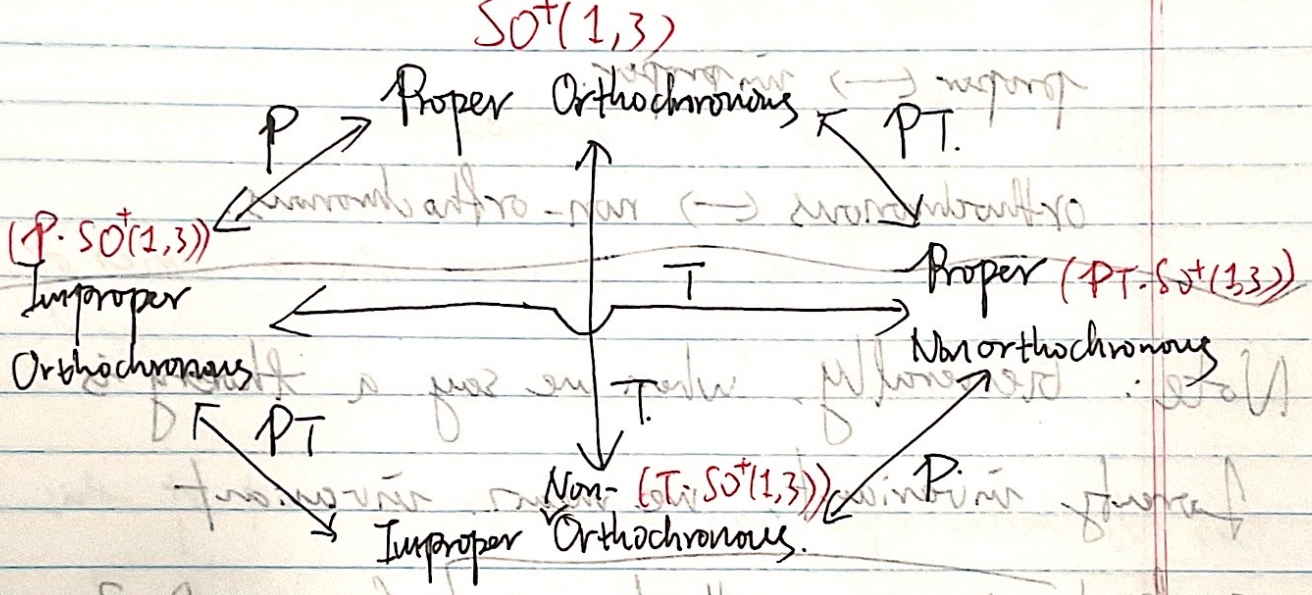
is L.T. invariant in the sense

$$(-\bar{\partial} + m^2/\hbar^2 c^2)\underbrace{\varphi(\Lambda^{-1}x)}_{\text{new field}} = 0. \quad (2)$$

Comment on passive & active transformations (possibly also Target Manifold & Embedding manifold)

1 Not just by analogy. A scalar has $U(\Lambda)^{-1}\varphi(0)U(\Lambda) = \varphi(0)$: no index, and Λ fixes the origin. Then p. 11 and (2.15) give $U^{-1}\varphi(x)U = e^{-i(\Lambda P)\cdot x/\hbar}\varphi(0)e^{i(\Lambda P)\cdot x/\hbar} = \varphi(\Lambda^{-1}x)$, as $(\Lambda P)\cdot x = P\cdot(\Lambda^{-1}x)$. Active vs passive: FieldTransformations.

2 Should read $(-\bar{\partial}^2 + m^2c^2/\hbar^2)\varphi(\Lambda^{-1}x) = 0$. Here $\bar{\partial}^\nu \equiv \partial/\partial\bar{x}_\nu$ acts on the argument $\bar{x} = \Lambda^{-1}x$; by the chain rule $\partial^\mu = \Lambda^\mu{}_\nu\bar{\partial}^\nu$, the rule above.



These are the four "Connected Component" (topological concept)